UNITED STATES INTERNATIONAL UNIVERSITY – AFRICA

School of Science and Technology

Final Group Project

DSA 3050 – Business Intelligence & Data Visualization

Education Analytics:

Student Performance & Retention Analytics Report

Open University Learning Analytics Dataset (OULAD)

Group F

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1. Executive Summary

> **Business Question:** *Are we meeting performance and retention targets, and where are the earliest warning signs of student failure?*

Figure 1: Executive Summary Dashboard — KPIs, Score Trends & Submission Activity

This report analyses student performance, engagement, and retention using the Open University Learning Analytics Dataset (OULAD), covering 32,593 students across 22 courses with over 10.6 million VLE interaction entries.

Four KPIs anchor the performance health check:

- **Assessment Pass Rate:** 72% of students who submit assessments score 40 or above.
- **Completion Rate:** Only 36% of enrolled students reach a passing or distinction final result.
- **Dropout Rate:** 26% of students formally withdraw before course completion.
- **At-Risk Students:** 3,824 students flagged as at-risk based on average score and engagement patterns.

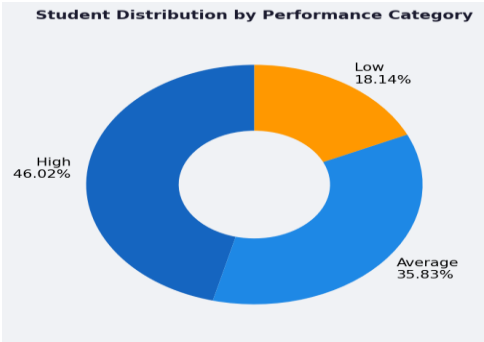

Figure 2: Student Distribution by Performance Category — High 46.02%, Average 35.83%, Low 18.14%

The performance distribution shows only 46.02% of students achieve High performance, 35.83% are Average, and 18.14% are Low. Average scores start near 80 at the beginning of the academic period and decline to approximately 70 by the end. Assessment submissions peak in January at approximately 16,600 and fall to fewer than 2,000 by November, reflecting the scale of dropout across the academic year.

2. Student Demographics & Segmentation

> **Business Question:** *Who is enrolled, and how do age, education background, gender, and region influence academic outcomes?*

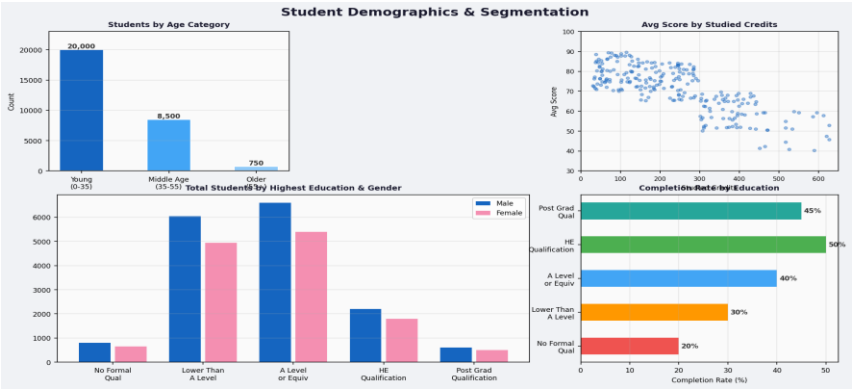

Figure 3: Student Demographics Dashboard — Age, Education, Gender & Completion by Education Level

2.1 Age Distribution

Young students (0–35 years) dominate enrolment at approximately 20,000 students, roughly 70% of the total cohort. Middle-aged students (35–55) make up around 8,500, and older students (55+) represent only approximately 750 students. Young students are the largest group enrolled, yet older students are the highest performers.

2.2 Education Level & Gender

A Level or Equivalent students form the largest enrolment group at approximately 12,000 students, followed closely by Lower Than A Level at approximately 11,000. HE Qualification holders account for around 4,000 students, with No Formal Qualifications and Post Graduate the smallest groups. Gender is broadly balanced across all education bands, with males slightly dominant in most categories.

2.3 Completion Rate by Education Level

The completion rate shows a clear positive gradient directly linked to entry qualifications:

- **HE Qualification:** ~50% completion. Students already experienced in higher education navigate academic expectations more effectively.
- **A Level or Equivalent:** ~40% completion. Traditional academic preparation provides a solid foundation.
- **Lower Than A Level:** ~30% completion. A significant gap requiring additional academic support.
- **No Formal Qualifications:** ~20% completion. Less than half the success rate of HE-qualified students, making this group the highest priority for additional support.
- **Post Graduate Qualification:** ~45%. Slightly below HE, possibly due to more complex work and life commitments.

2.4 Credits vs. Academic Performance

The scatter chart confirms a clear negative correlation between studied credits and average score. Students with 0–150 credits cluster between 70–90 in score. At 150–300 credits, scores remain 65–85 but variability increases. At 300–450 credits, scores drop to the 50–70 range. Students

carrying 450–600+ credits score as low as 40–60. This points to the need for credit caps or mandatory academic advising for students exceeding 300 studied credits.

3. Geographic Distribution & Digital Engagement

> **Business Question:** *Where are our students, which digital resources are they using, and do gender and age create structural differences in engagement and completion?*

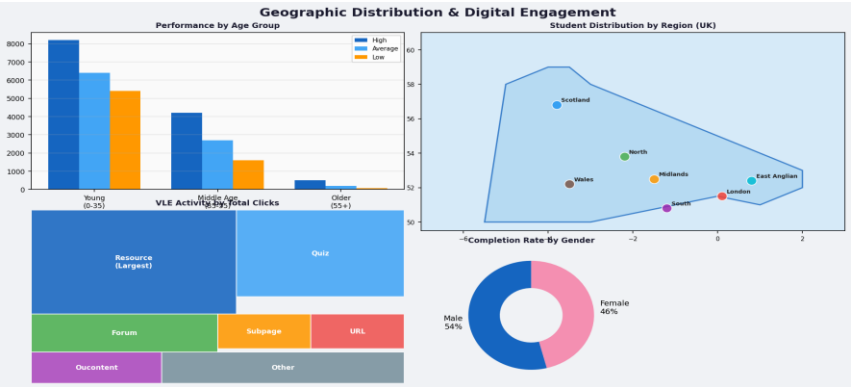

Figure 4: Geographic & Engagement Dashboard — Region Map, VLE Activity & Performance by Age

3.1 Geographic Reach

The geographic map shows students distributed across all UK regions, with the highest concentrations in the Midlands, London, and South. East Anglia and Wales have notably lower enrolment volumes, consistent with the drill-through analysis that flags East Anglia as a region requiring targeted retention attention. For a distance-learning institution, this distribution has direct implications for support service alignment and connectivity assumptions.

3.2 VLE Activity Engagement

The VLE treemap shows Resource materials generating by far the highest total clicks, followed by Quiz interactions and Forum activity. Resource and quiz engagement are the most meaningful indicators of active learning. Low-engagement activity types such as URLs and Oucontent should be reviewed. Published OULAD research confirms VLE engagement is the single strongest predictor of student outcome, achieving AUC scores of up to 0.91 for dropout prediction when using VLE data alone.

3.3 Performance by Age Group

The clustered bar chart shows older students have a higher rate of High performance relative to their small enrolment. Young students (20,000 enrolled) make up the highest counts across all performance categories including Low and Withdrawn. Middle-aged students show a better performance distribution than the young cohort.

3.4 Completion Rate by Gender

The donut chart shows male students account for approximately 54% of completions versus 46% for female students. This gap should be assessed against enrolment proportions to determine whether it reflects a genuine outcome disparity requiring targeted support.

4. At-Risk Student Deep Dive

> **Business Question:** *For a specific student segment or region, who is at risk, when are they disengaging, and what would a targeted intervention realistically achieve?*

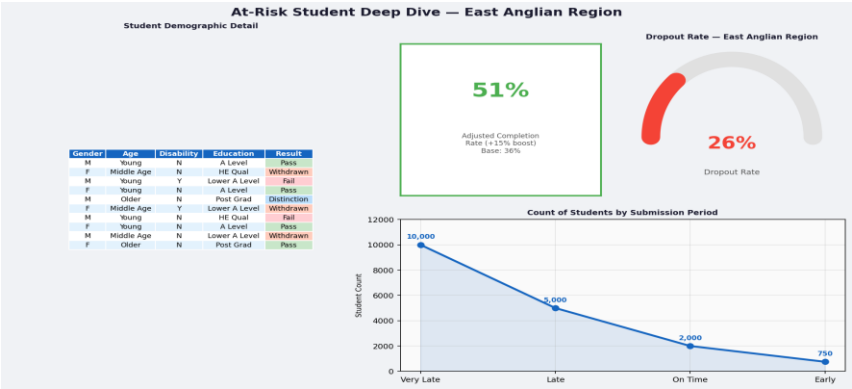

Figure 5: Drill-Through Dashboard — East Anglian Region: Demographics, Submissions, Dropout & Adjusted Completion

4.1 Student Profile in East Anglia

The demographic detail table shows the profile of students in the East Anglian Region, including gender, age category, disability status, highest education level, and final result. Withdrawn students in this segment tend to be middle-aged with lower entry qualifications and disability flags, consistent with the finding that prior education and disability status are key risk factors.

4.2 Submission Timing: The Early Warning Signal

The submission period chart shows a clear pattern:

- **Very Late Submissions:** ~10,000 students, representing 35–40% of total enrolment.
- **Late Submissions:** ~5,000 students.
- **On-Time Submissions:** ~2,000 students.
- **Early Submissions:** ~500–750 students, the smallest group by far.

This pattern is institution-wide, not an outlier. Students submitting very late are highly likely to Fail or Withdraw. With 10,000 students submitting very late, late submission is the norm, not the exception.

4.3 Dropout Rate & Intervention Modelling

The dropout rate gauge confirms 26% of students withdraw. The adjusted completion rate card models the impact of a 15% targeted intervention boost applied to the current 36% completion baseline, projecting an adjusted completion rate of approximately 51%. A well-designed early alert programme does not need to be large-scale to make a difference — a 15% uplift would move over 4,800 additional students into completion.

5. Key Influencers of Student Outcomes

> **Business Question:** *What factors most significantly drive student completion or dropout, and which student profiles should receive priority intervention?*

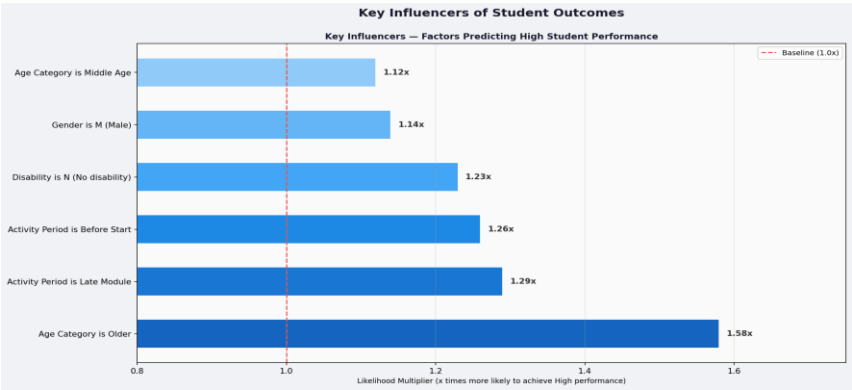

Figure 6: Key Influencers — Factors Predicting High Student Performance (Power BI AI Visual)

The Key Influencers visual uses Power BI's analysis to identify which variables most strongly predict whether a student achieves High performance. Results are expressed as likelihood multipliers — how many times more likely a student with that characteristic is to achieve High performance compared to the average.

5.1 Age: The Strongest Predictor

Age Category being Older is the strongest predictor at 1.58x — older students are 58% more likely to achieve High performance than the average student. Age Category being Middle Age also shows higher performance at 1.12x. This is at odds with the enrolment pattern, where young students make up 20,000 enrolled compared to only 750 older students.

5.2 Engagement Timing Matters

Activity Period being Late Module predicts High performance at 1.29x, and Activity Period being Before Start at 1.26x. Students who engage with VLE materials before the official module start date are 26% more likely to succeed. This makes pre-module orientation and early access to materials a practical and low-cost intervention.

5.3 Disability and Gender Gaps

Students without disability (Disability = N) are 1.23x more likely to achieve High performance, indicating that current support systems are not fully meeting the needs of students with disabilities. Male students are 1.14x more likely to achieve High performance than female students — a smaller gap but one worth investigating in the context of course design and assessment methods.

6. Summary of Findings & Recommended Actions

The five analytical sections of this report answer a connected chain of institutional questions, from the broad to the specific, grounded in actual dashboard values from the OULAD Power BI analysis.

6.1 Treat the Submission Crisis as a Systemic Problem

With ~10,000 students submitting very late (35–40% of total enrolment), late submission is not an individual issue but an institutional one. An automated early alert system triggered by late or missing submissions, with personal tutor outreach, is the single highest-impact intervention available. Modelling shows a 15% improvement in completion would add approximately 4,800 completions to the current 36% baseline.

6.2 Redesign VLE Engagement Strategy

VLE interaction is the strongest statistical predictor of student success. Offering pre-module access to materials could boost "Before Start" engagement, currently a 1.26x predictor of High performance. Course design reviews should prioritise resource and quiz-type materials, which generate the highest engagement, and reduce or restructure low-click activity types.

6.3 Introduce Credit Load Guidance

Students carrying 300+ credits score 50–70 on average versus 70–90 for students with under 150 credits. Academic advisors should use credit thresholds to guide students toward manageable study loads before enrolment.

6.4 Foundation Support for Lower-Qualified Entrants

The completion rate gap between No Formal Qualifications (~20%) and HE Qualification (~50%) is 30 percentage points. With approximately 11,000 students entering with Lower Than A Level qualifications — the second-largest group — even modest improvements in this segment's completion rate would have a measurable institutional impact.

6.5 Address the Disability & Gender Gaps

Students without disabilities are 1.23x more likely to succeed and male students 1.14x more likely. An accessibility audit of course delivery, assessment formats, and support mechanisms is warranted. Gender gaps should be investigated in the context of assessment design and learning style alignment.

6.6 Regional Focus: East Anglia and Lower-Enrolment Regions

The drill-through analysis for the East Anglian Region shows higher dropout relative to larger regions. Lower-enrolment regions may have fewer peer support networks. Regional retention

strategies, including local tutoring partnerships and adjusted completion targets, should be considered.

Data Source: *https://www.kaggle.com/datasets/anlgrbz/student-demographics-online-education-dataoulad*